

Medical Inventory Analysis Report

Project Title: Medical Inventory Optimization & Stock Efficiency Analysis

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1. Executive Summary

This project focuses on analysing medical inventory data to identify stock inefficiencies, supplier delays, overstock situations, and out-of-stock (OOS) risks. The analysis was performed using a dataset containing over 14,000 rows of inventory records.

The main objective of the project was to improve inventory efficiency, reduce unnecessary storage costs, and support better stock planning decisions through data analysis and dashboard reporting.

Key Highlights

- Analysed 14,000+ rows of medical inventory data
 - Identified recurring stockouts, supplier delays, and slow-moving inventory
 - Found that 31% of medicines were overstocked, impacting storage utilization and profitability
 - Created custom business indicators such as Overstock Flag, Slow-Moving Indicator, and Return Ratio
 - Built an interactive Power BI dashboard to monitor stock efficiency and supplier performance
 - Recommended optimized reorder strategies and clearance planning for slow-moving inventory
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2. Data Overview

Data Source

Medical inventory and supplier transaction dataset collected from internal business records.

Data Preparation Included

- Cleaning missing and inconsistent values
- Standardizing medicine names and supplier information
- Removing duplicate records
- Creating calculated business indicators for inventory analysis
- Transforming raw inventory data into analysis-ready format using SQL, Python, and Power BI

Final Dataset Size

- 14,000+ inventory records

Key Variables Analysed

- Medicine Category
 - Stock Quantity
 - Supplier Lead Time
 - Return Quantity
 - Overstock Status
 - Slow-Moving Inventory
 - Out-of-Stock (OOS) Frequency
 - Supplier Performance Metrics
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3. Dashboard Insights

Overstock Analysis

- 31% of medicines were identified as overstocked
- Overstocked items occupied unnecessary storage space and increased holding costs

Insight

Excess inventory directly affected storage efficiency and profitability. Better demand forecasting and reorder planning can reduce overstock situations.

Slow-Moving Inventory

- Multiple products showed low sales movement over extended periods
- Slow-moving items contributed to inventory blockage and reduced cash flow efficiency

Insight

Certain medicines had consistently low demand and required clearance strategies or reduced reorder frequency.

Supplier Performance Analysis

- Several suppliers showed repeated delivery delays
- Delayed deliveries contributed to recurring stockouts for important medicines

Insight

Supplier reliability has a direct impact on inventory availability. Monitoring supplier lead times can improve procurement planning.

Stockout (OOS) Trends

- High-demand medicines experienced recurring out-of-stock situations
- OOS events impacted medicine availability and operational efficiency

Insight

Improper reorder timing and supplier delays were major contributors to stockout issues.

4. Business Impact

- Reduced operational inefficiencies caused by excess inventory
 - Improved visibility into supplier performance and stock movement
 - Helped identify high-risk inventory categories requiring immediate attention
 - Supported better inventory planning and stock optimization decisions
 - Enabled management to monitor inventory KPIs through an interactive Power BI dashboard
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5. Recommendations

Short Term

- Reduce reorder quantities for overstocked medicines
- Create clearance or promotional strategies for slow-moving inventory
- Monitor critical medicines with frequent stockouts more closely
- Improve tracking of supplier delivery timelines

Medium Term

- Implement optimized reorder point calculations based on demand patterns
- Categorize medicines based on movement and business priority
- Introduce supplier performance scorecards for procurement evaluation

Long Term

- Build predictive inventory forecasting models using historical demand data
 - Automate inventory alerts for overstock and stockout risks
 - Improve supply chain coordination to reduce delivery delays and inventory imbalance
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6. Conclusion

This project demonstrates how data analysis and dashboard reporting can improve medical inventory management. The findings highlighted major challenges such as overstocking, slow-moving inventory, supplier delays, and recurring stockouts.

By implementing optimized reorder strategies, improving supplier monitoring, and using data-driven inventory planning, businesses can reduce storage costs, improve medicine availability, and increase overall operational efficiency.